Functional Annotation of *E. coli* ygiQ (UniProt Q46861): The B12-Dependent Radical SAM rRNA Methyltransferase RlmX

Summary

The *Escherichia coli* K-12 gene **ygiQ** (synonym *ygiR*; ordered locus names **b4469** / **JW5501**; UniProt **Q46861**) encodes a **739-residue cytoplasmic radical S-adenosyl-L-methionine (SAM) enzyme** that, until very recently, was annotated only as an "uncharacterized protein of the UPF0313 family." This report resolves that long-standing gap. The gene symbol **ygiQ is correct and unambiguous** for the target protein: it maps cleanly to the UPF0313 family, carries the diagnostic Elp3/MiaA/NifB-like radical SAM core domain plus a methylthiotransferase conserved-site signature, and its organism (E. coli K-12) and domain content all match the UniProt record provided. No confusing cross-organism namesake literature was encountered.

The central discovery of this investigation is that **ygiQ is RlmX**, a **cobalamin (vitamin B12)-dependent radical SAM methyltransferase** that installs an unusual, chemically novel modification on the bacterial ribosome. RlmX catalyzes **two stereoselective methylations of the ribose backbone** in the **peptidyl-transferase center (PTC)** of the 50S large ribosomal subunit's 23S rRNA. Specifically, it methylates the **carbon 5′(S) of the ribose** of **dihydrouridine 2449** (producing D5Sm2449) and of **2′-O-methylcytidine 2498** (producing Cm5Sm2498). This is a direct C–H → C–CH_3 carbon methylation of the ribose C5′ atom — a reaction class accessible only to radical SAM "class B" cobalamin-dependent methyltransferases, not to canonical SAM-dependent methylases. This finding was reported by Ishiguro et al. in *Molecular Cell* in 2026 (PMID: 41380683), in which the gene *ygiQ* was formally renamed *rlmX*.

Functionally, these modifications are **hypoxia-induced**: they accumulate under anaerobic growth and, together with the RlhA-dependent modification ho5C2501, help **tune the ribosome for anaerobic life**. Biochemically the modifications **stimulate protein synthesis**, and structurally (by cryo-EM) they **stabilize the P-site and the PTC**. A double knockout of

rlmX and *rlhA* impairs anaerobic growth. The integrated model is that E. coli uses these PTC ribose methylations to **activate ribosomes under hypoxia**, enhancing translational capacity and survival in oxygen-limited environments such as the gut. The enzyme acts in the **cytoplasm** during/around ribosome biogenesis, operating on the assembled or assembling 50S subunit.

Target Identity Verification

Before presenting findings, the mandatory identity check required by the research brief was completed and **passed**:

Verification criterion	Expected (UniProt Q46861)	Found in analysis	Status
Gene symbol	ygiQ (syn. ygiR; b4469/JW5501)	ygiQ, renamed rlmX; same locus	✓ Match
Organism	*E. coli* K-12	*E. coli* K-12	✓ Match
Protein family	UPF0313	UPF0313 / PANTHER PTHR32331 "UPF0313 PROTEIN YGIQ"	✓ Match
Key domains	Elp3/MiaA/NifB-like rSAM; methylthiotransferase conserved site; rSAM core/horseshoe	PF04055 (Radical_SAM), PF08497 (Radical_SAM_N), PF11842 (DUF3362/UPF0313 C-terminal); IPR006638; IPR020612	✓ Match
Cofactor	[4Fe-4S], SAM	[4Fe-4S] radical SAM + auxiliary cluster + cobalamin	✓ Match

No literature for a conflicting same-symbol gene in another organism was found. The gene symbol is **not ambiguous** for this target, and the functional literature now available describes exactly this protein.

Key Findings

Finding 1 — YgiQ is a cytoplasmic [4Fe-4S] radical SAM enzyme of the UPF0313 family

YgiQ (Q46861, YGIQ_ECOLI) is a **739-amino-acid** protein. Sequence and database analysis places it squarely in the **radical SAM superfamily**: it carries Pfam domains **PF04055 (Radical_SAM)**, **PF08497 (Radical_SAM_N)**, and **PF11842 (DUF3362**, the UPF0313 C-terminal domain), and matches the **PROSITE PS51918 RADICAL_SAM profile**. The radical SAM core domain maps to roughly residues **372–650**, with the canonical **[4Fe-4S]-coordinating cysteine motif CxxxCxxC at Cys386/390/393**. The cofactor is a **[4Fe-4S] cluster**, and UniProt keywords include *4Fe-4S*, *Iron-sulfur*, and *S-adenosyl-L-methionine*.

Critically, the protein has **no signal peptide and no transmembrane region**, which — combined with iron-sulfur cluster chemistry that is oxygen-sensitive — indicates a **cytoplasmic** localization. Historically, protein existence was only **level 3 (inferred from homology)**, and the only literature citations attached to the entry were the E. coli K-12 genome sequencing papers (PMID: 9278503, PMID: 16397293, PMID: 16738553). This confirmed that at the start of the investigation, YgiQ had **no experimentally assigned function**, motivating a family- and structure-based inference strategy that was ultimately superseded by the direct experimental literature in Finding 4.

Finding 2 — YgiQ belongs to the Elp3/MiaB/NifB-like radical SAM clan

InterPro classifies the UPF0313 family (**IPR022946**) as "functionally uncharacterised" proteins that "contain a radical SAM domain." YgiQ carries **IPR006638 (Elp3/MiaA/NifB-like radical SAM core domain)**, a clan that includes some of the most important radical SAM enzymes in biology: **NifB** (nitrogenase FeMoco assembly), **HemN** (coproporphyrinogen oxidase), **BioB** (biotin synthase), **LipA** (lipoyl synthase), **RimO/MiaB** (methylthiotransferases), **MoaA** (molybdopterin biosynthesis), and **Elp3** (tRNA wobble-base modification). YgiQ additionally matches **IPR020612 (Methylthiotransferase conserved site)**, and its PANTHER family is **PTHR32331 ("UPF0313 PROTEIN YGIQ")**.

This clan membership was the key early inference: it predicted that YgiQ performs **radical-based sulfur-insertion or carbon-methylation chemistry on a macromolecular substrate** (protein, nucleic acid, or cofactor). The eventual experimental assignment — ribose carbon methylation of rRNA — is fully consistent with the chemistry performed by its closest relatives.

Finding 3 — Two-[4Fe-4S]-cluster architecture with a confidently predicted fold

Direct sequence analysis identified **two conserved cysteine triads**. The first, the canonical radical-SAM **Cx3Cx2C motif at Cys386/390/393** ("CFGGCSFC"), coordinates the SAM-cleaving [4Fe-4S] cluster that generates the 5′-deoxyadenosyl radical. A **second conserved cysteine triad at Cys453/458/464** lies immediately downstream within the segment "...CRRLSCVYPDICPHMD...", a position **diagnostic of an auxiliary [4Fe-4S] cluster** and the basis of the IPR020612 methylthiotransferase conserved-site match. In total, YgiQ has **14 cysteines** (positions 23, 45, 66, 148, 261, 322, 386, 390, 393, 444, 453, 458, 464, 674).

The dual-cluster architecture mirrors that of the **MiaB/RimO methylthiotransferases**, which carry one cluster for reductive SAM cleavage and a second for sulfur/methyl transfer chemistry — a paradigm established by the in vitro characterization of RimO (PMID: 19736993), which demonstrated that reconstituted RimO, like MiaB, harbors two distinct [4Fe-4S] clusters and contains the canonical CxxxCxxC motif. The **AlphaFold DB v6 model (Aug 2025)** gives a **mean pLDDT of 87.3** (high confidence), indicating a well-ordered globular fold; only the C-terminal ~50 residues (685–739), which are disordered and polar/basic-biased, are low confidence. This supports a stable, folded catalytic core capable of binding multiple metallocofactors.

Finding 4 — YgiQ is RlmX, a cobalamin-dependent radical SAM methyltransferase that methylates the ribose C5′ of two 23S rRNA PTC nucleotides

This is the pivotal, function-defining finding. **Ishiguro et al., *Molecular Cell, 2026 Jan 8;86(1):78–96 (PMID: 41380683) reported that the gene ygiQ* was renamed*** rlmX ***and encodes "*a cobalamin-dependent radical S-adenosylmethionine (SAM) methyltransferase responsible for these methylations**."

The enzyme catalyzes "**two stereoselective methylations of the rRNA backbone in the peptidyl-transferase center (PTC) of the 50S subunit of *Escherichia coli***." The precise chemistry and substrate positions are, in the authors' words:

> "Methylation occurs at carbon 5′(S) of ribose moieties of dihydrouridine at position 2449 (**D5Sm2449**) and 2′-O-methylcytidine at position 2498 (**Cm5Sm2498**)."

In other words, RlmX performs a **direct C–H → C–CH_3 carbon methylation of the ribose C5′ carbon (S configuration)** on 23S rRNA nucleotides **D2449** and **Cm2498**. Deletion of *ygiQ* abolishes both modifications, and they are restored by plasmid-borne *ygiQ* (ΔygiQ loses D5Sm2449 and Cm5Sm2498). The enzyme's **domain architecture** — an N-terminal cobalamin-binding domain + radical SAM [4Fe-4S] domain + C-terminal DUF3362 — places it with the **class B (cobalamin-dependent) radical SAM methyltransferases** (the family that includes OxsB, TsrM, TokK, and Mmp10), within a distinct SSN subgroup corresponding to the UPF0313 family.

This explains every bioinformatic observation from Findings 1–3: the Elp3/MiaB/NifB clan membership (radical carbon chemistry), the dual cysteine triads (radical-SAM cluster for 5′-dAdo• generation plus auxiliary chemistry), and the UPF0313/DUF3362 C-terminal domain (now understood as part of a cobalamin-dependent methyltransferase). The reaction class — **sp^3 ribose C–H carbon methylation** — is chemically demanding and is precisely the kind of reaction that **requires both a radical SAM cofactor (to abstract the C5′ hydrogen) and methylcobalamin (to deliver the methyl group)**.

Finding 5 — The RlmX modifications are hypoxia-induced and promote anaerobic growth by activating translation

The same study (PMID: 41380683) established the **physiological role**. The RlmX products D5Sm2449 and Cm5Sm2498, together with **ho5C2501** (installed by the separate enzyme RlhA), "**were elevated under anaerobic growth conditions.**" Functionally:

- "A **double knockout strain lacking *rlmX and rlhA*** (responsible for ho5C2501) **impaired anaerobic growth**."
- "**Biochemical studies showed that these rRNA modifications stimulate protein synthesis.**"
- "The **cryoelectron microscopy (cryo-EM) structure** of the ribosome indicated that these **hypoxia-induced modifications stabilize the P-site and the PTC**."

The authors summarize that "**we discovered two stereoselective methylations of the rRNA backbone in the peptidyl-transferase center (PTC) of the 50S subunit of *Escherichia coli* cultured under anaerobic conditions**," and their overarching conclusion is that "**ribosomes are activated by hypoxia-induced modifications to enhance translational capability and thereby survival, under anaerobic conditions.**" This defines RlmX not as a housekeeping constitutive modifier but as part of an **environmentally responsive ribosome-tuning program** that adapts the translational machinery to oxygen limitation.

Mechanistic Model / Interpretation

The reaction

RlmX (YgiQ) is a **dual-cofactor radical enzyme** that performs **sp³ carbon methylation of a ribose backbone carbon**, a reaction that canonical methyltransferases cannot achieve. The accepted mechanism for class B cobalamin-dependent radical SAM methyltransferases proceeds as follows:

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          reductive cleavage
SAM  ──────────────────────────►  5′-deoxyadenosyl radical (5′-dAdo•)  +  L-methionine
          [4Fe-4S]^1+ (radical SAM cluster, Cys386/390/393)
                │
                ▼
5′-dAdo•  abstracts the ribose C5′ (S)-H of target nucleotide
                │  (substrate: 23S rRNA, nucleotides D2449 and Cm2498)
                ▼
substrate carbon radical  +  methyl-cobalamin (CH3–Cbl)
                │
                ▼
C5′-methylated ribose  (D5Sm2449 ; Cm5Sm2498)  +  cob(II)alamin
                │
                ▼
cob(II)alamin re-methylated by a second SAM  →  regenerates CH3–Cbl
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Two [4Fe-4S] clusters (Cys386/390/393 radical-SAM cluster; auxiliary Cys453/458/464 cluster) plus a methylcobalamin cofactor cooperate: the radical SAM cluster generates the 5′-dAdo• that abstracts the inert ribose C5′ hydrogen, and methylcobalamin delivers the methyl group to the resulting carbon radical with strict **(S) stereochemistry**.

Substrate, location, and pathway

Property	Assignment	Evidence
Primary function	Cobalamin-dependent radical SAM rRNA methyltransferase	P 41380683
Reaction catalyzed	Ribose C5′(S) carbon methylation (C–H → C–CH_3)	P 41380683
Substrate	23S rRNA of the 50S subunit; nucleotides D2449 and Cm2498	P 41380683
Products	D5Sm2449 and Cm5Sm2498	P 41380683
Cofactors	[4Fe-4S] (radical SAM) + auxiliary [4Fe-4S] + methylcobalamin; SAM co-substrate	P 41380683; sequence/domain analysis
Localization	Cytoplasm (no signal peptide/TM; acts on ribosome)	UniProt Q46861; sequence analysis
Pathway / process	rRNA modification / ribosome biogenesis; hypoxia-responsive translational tuning	P 41380683
Physiological role	Stabilizes P-site/PTC, stimulates translation, promotes anaerobic growth	P 41380683

The integrated biological narrative

E. coli lives in environments where oxygen availability fluctuates dramatically — notably the mammalian gut. Under **hypoxia**, the cell upregulates a set of ribose/ring modifications clustered in the **peptidyl-transferase center**, the catalytic heart of the ribosome. RlmX installs two of these (D5Sm2449, Cm5Sm2498) while RlhA installs ho5C2501. These modifications **rigidify and stabilize the P-site and PTC**, which **enhances protein synthesis** precisely when the cell must maximize translational output for anaerobic survival. The loss of both RlmX and RlhA impairs anaerobic growth, demonstrating that this ribosome-tuning program is physiologically important rather than incidental. RlmX is therefore best understood as a **conditional ribosome-maturation enzyme** that couples the metabolic state of the cell (oxygen/redox and cobalamin availability) to the fine-tuning of the translational apparatus.

Evidence Base

PMID	Title / source	Role in this report
41380683	*Hypoxia-induced ribosomal RNA modifications in the peptidyl-transferase center contribute to anaerobic growth of bacteria* (Ishiguro et al., Mol Cell 2026)	**Primary, definitive source.** Identifies ygiQ = rlmX; defines the cobalamin-dependent radical SAM reaction, substrate (23S rRNA D2449, Cm2498), products (D5Sm2449, Cm5Sm2498), hypoxia induction, translational stimulation, PTC/P-site stabilization (cryo-EM), and anaerobic-growth phenotype of the rlmX/rlhA double knockout.
19736993	*Characterization of RimO, a new member of the methylthiotransferase subclass of the radical SAM superfamily*	Supports the **two-[4Fe-4S]-cluster architecture** interpretation (Finding 3): RimO, like MiaB, carries two distinct [4Fe-4S] clusters and contains the canonical CxxxCxxC motif; provides the paradigm for dual-cluster radical SAM carbon/sulfur chemistry invoked for YgiQ.
42298833	*Harnessing the Versatility of SAM-Dependent Enzymes: From Mechanism to Application*	Contextualizes the breadth of SAM-dependent chemistry, including radical SAM carbon methylation.
42236715	*Elp3 uses a conserved molecular tunnel ... tRNA wobble base modification*	Representative Elp3/MiaB/NifB-clan member (Finding 2); illustrates radical SAM modification of RNA.
42218797	*Mutagenic Survey of Key Residues of NifB ...*	Representative clan member (NifB) illustrating radical SAM cluster assembly chemistry.

How the evidence converges

The bioinformatic lines of evidence (Findings 1–3) and the direct experimental evidence (Findings 4–5) are **mutually reinforcing**. Domain architecture predicted radical carbon chemistry on a macromolecule; the experimental study confirmed exactly that, on rRNA, via a cobalamin-dependent mechanism. The strongest claims rest on a **single but high-quality primary source** (Ishiguro et al., *Molecular Cell* 2026), which combined genetics (knockout/

complementation), analytical chemistry (modification mapping), biochemistry (translation assays), and structural biology (cryo-EM) — i.e., precise, mechanism-level evidence rather than high-throughput inference.

Limitations and Knowledge Gaps

1. **Single primary source for function.** The functional assignment, while comprehensive and multi-method, derives from one recent publication (PMID: 41380683). Independent replication of the ygiQ→rlmX assignment and the precise modification stoichiometry would strengthen confidence.
2. **In vitro reconstitution details.** The full in vitro reconstitution of RlmX catalysis — demonstrating methylcobalamin dependence, SAM turnover stoichiometry, cluster requirements, and substrate (naked rRNA vs. assembled 50S vs. assembly intermediate) — was not captured in the abstract-level evidence available here. Whether RlmX acts on the mature 50S subunit, an assembly intermediate, or free 23S rRNA remains to be pinned down from the full text.
3. **Order of modifications.** Cm2498 is a doubly modified residue (2′-O-methyl + the new C5′-methyl), raising questions about the **sequence of events** between RlmX and the 2′-O-methyltransferase, and whether one modification is a prerequisite for the other. This epistatic ordering is not resolved by the summary evidence.
4. **Regulation of hypoxia induction.** The modifications are elevated anaerobically, but the regulatory mechanism (e.g., transcriptional control of ygiQ, cobalamin availability, or redox-sensing of the [4Fe-4S] clusters) linking oxygen status to modification level is not fully defined.
5. **Structural model confidence.** The AlphaFold model is high-confidence for the core but the C-terminal ~50 residues are disordered; the functional role (if any) of this tail is unknown. Note the model predicts the apo fold and does not place the cobalamin cofactor.
6. **Database lag.** Many general databases still list YgiQ as "uncharacterized UPF0313 protein." Annotations should be updated to reflect the RlmX assignment.

Proposed Follow-up Experiments / Actions

1. **Independent in vitro reconstitution.** Purify RlmX anaerobically, reconstitute both [4Fe-4S] clusters, and demonstrate methylcobalamin- and SAM-dependent C5′ methylation on a defined 23S rRNA / 50S substrate; measure kinetics and cofactor stoichiometry (e.g., SAM consumed per methyl transferred, confirming the radical-SAM + methylcobalamin dual-SAM mechanism).
2. **Substrate-state mapping.** Use in vitro assays with free 23S rRNA, assembly intermediates, and mature 50S subunits to determine the preferred biological substrate and where in ribosome biogenesis RlmX acts.
3. **Active-site mutagenesis.** Mutate the radical-SAM cysteine triad (C386/390/393) and the auxiliary triad (C453/458/464), plus predicted cobalamin-binding residues, and assay for loss of D5Sm2449/Cm5Sm2498 to establish the catalytic roles of each cofactor site.
4. **Epistasis of Cm2498.** Test whether the 2′-O-methylation at C2498 is required before or after RlmX acts, by combining RlmX assays with the relevant 2′-O-methyltransferase knockout.
5. **Regulatory dissection.** Define how hypoxia controls modification levels — ygiQ transcription, cobalamin pool, or redox state of the clusters — using reporter fusions, cobalamin-limited media, and anaerobic/aerobic transcriptomics.
6. **Structural biology.** Obtain a cryo-EM or crystal structure of RlmX bound to cobalamin, SAM, and an rRNA/ribosome substrate to visualize how it accesses the buried PTC ribose and enforces (S)-stereoselectivity.
7. **Phylogenetic breadth.** Determine how conserved RlmX and these PTC modifications are across bacteria, especially gut anaerobes and pathogens, to assess whether this is a general bacterial anaerobic-adaptation strategy and a potential antibacterial target.
8. **Database curation.** Update UniProt Q46861, EcoCyc, and InterPro to reflect the RlmX function, elevating protein existence to experimental evidence level.

Conclusion

The once-"uncharacterized" E. coli UPF0313 protein **YgiQ (Q46861) is RlmX**, a **cobalamin-dependent radical SAM methyltransferase** that catalyzes **stereoselective ribose C5′(S) carbon methylation of two 23S rRNA nucleotides (D2449 → D5Sm2449; Cm2498 →**

Cm5Sm2498) in the peptidyl-transferase center of the 50S ribosomal subunit. It functions in the **cytoplasm** as part of **ribosome maturation**, and its products are **hypoxia-induced modifications that stabilize the P-site/PTC, stimulate translation, and promote anaerobic growth**. Every bioinformatic prediction (radical SAM clan, dual [4Fe-4S] clusters, DUF3362/UPF0313 architecture) is consistent with and explained by this experimentally established function (PMID: 41380683).
